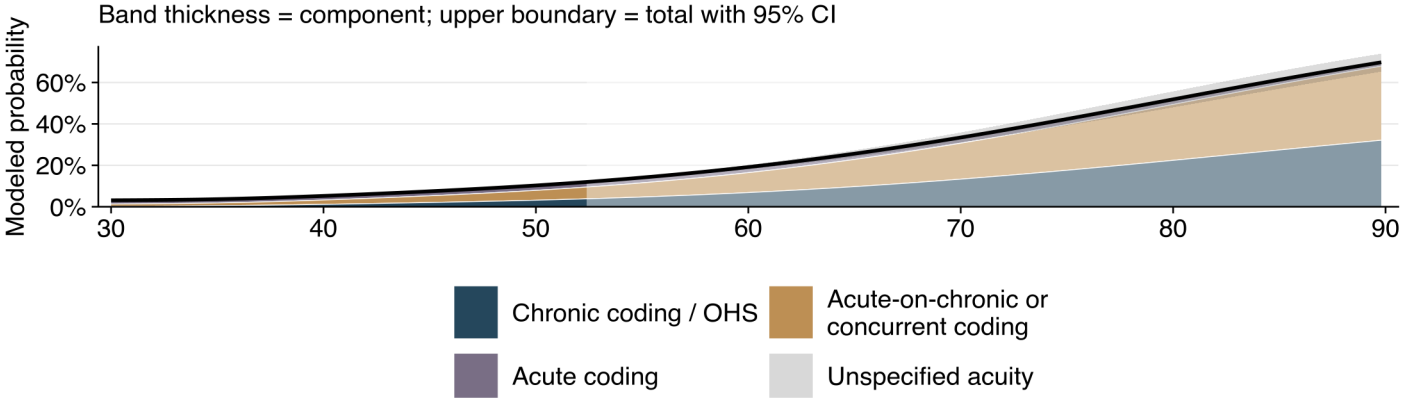
S2-ICD-acuity

BMI and documented respiratory-failure acuity / OHS

Primary population | Encounter-wide; model/reference N = 176,922 | Firth bias reduction
Events: chronic/OHS 1,769; acute/chronic 3,443; acute 3,294; unspecified 128

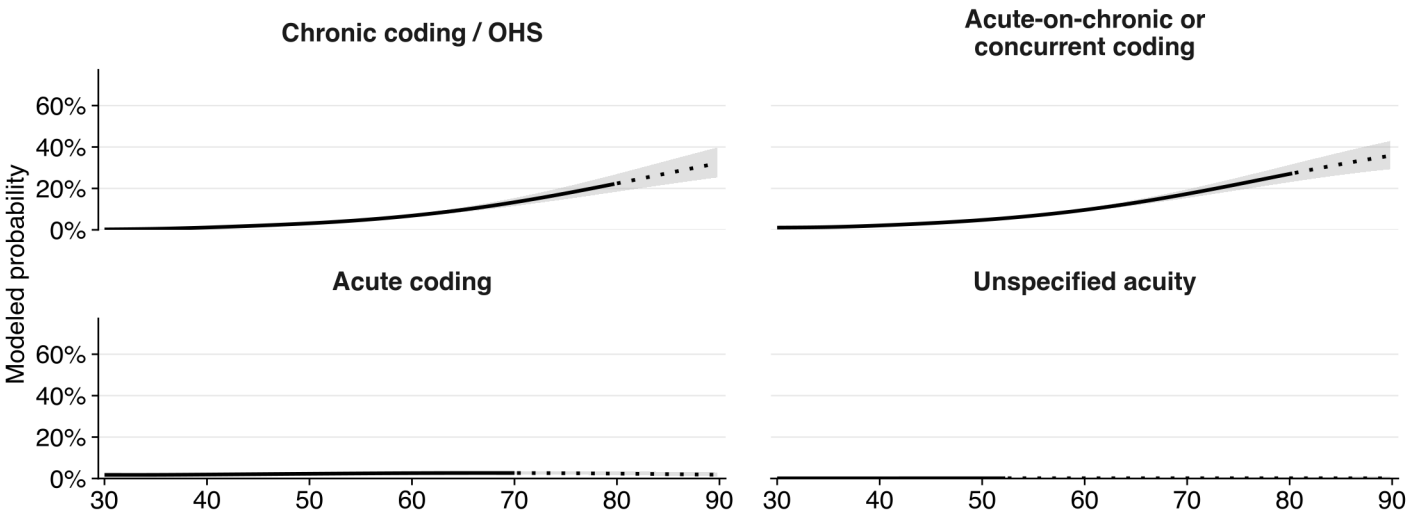
A Documented diagnoses divided by acuity coding

Band thickness = component; upper boundary = total with 95% CI



B Component magnitudes on common axes

Pointwise 95% intervals; dotted segments indicate local sparsity



C Patients in the shared diagnosis-capture population

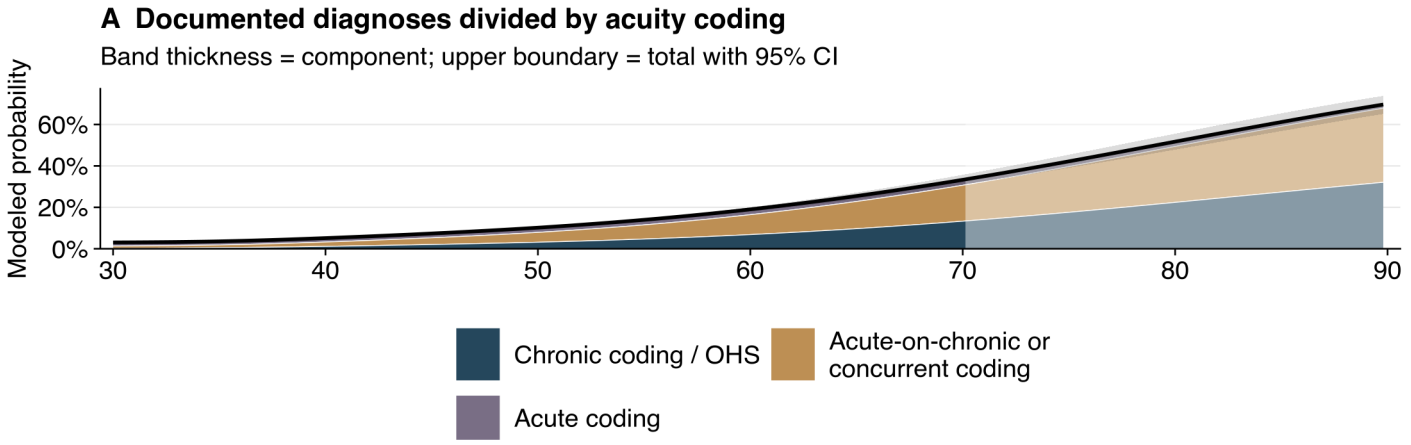


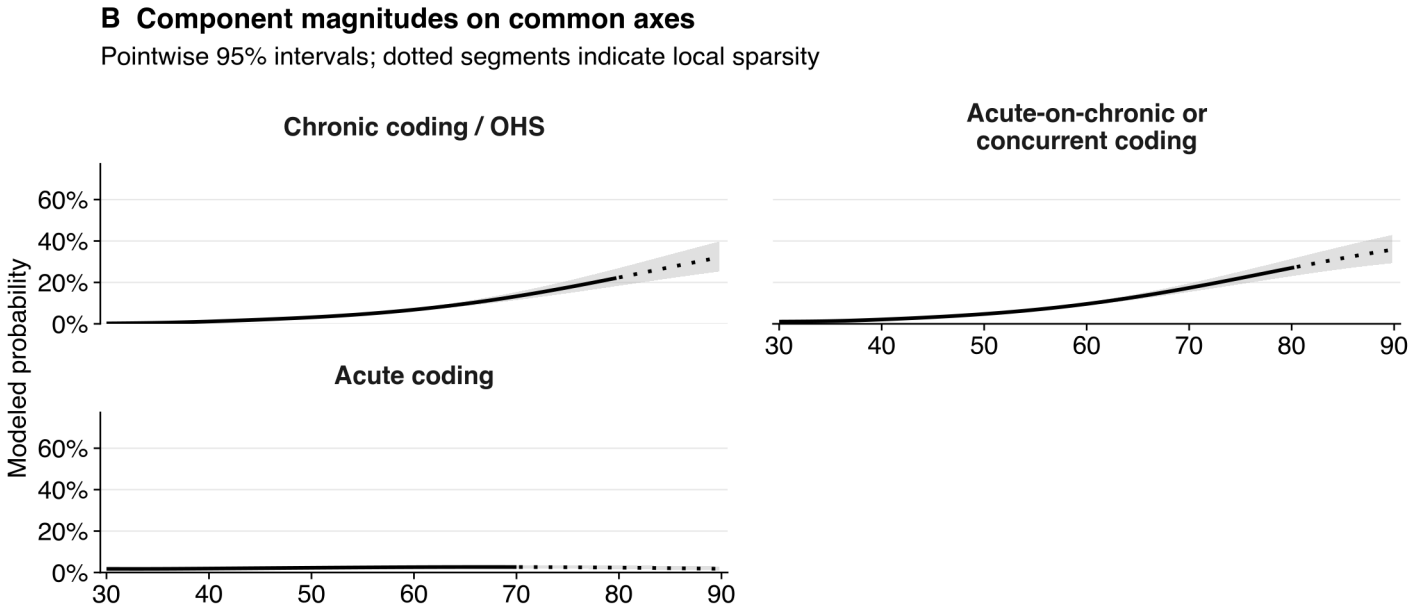
Mean-bias-reduced five-category multinomial spline; all diagnosis-evaluable primary patients retained. Encounter-wide coding, no gas requirement; inherited adjustment and primary spline basis. Bands partition the composite; thickness gives component probability. Shading: pointwise 95% intervals from joint information covariance, conditional on the empirical reference and basis. Fading/dots: local sparsity; histogram N equals model/reference N. Acute-on-chronic includes explicit or concurrent acute/chronic coding. OHS (obesity hypoventilation syndrome) is analytically chronic; this does not establish chronic respiratory failure. Source selection and diagnostic incorporation limit physiological interpretation.

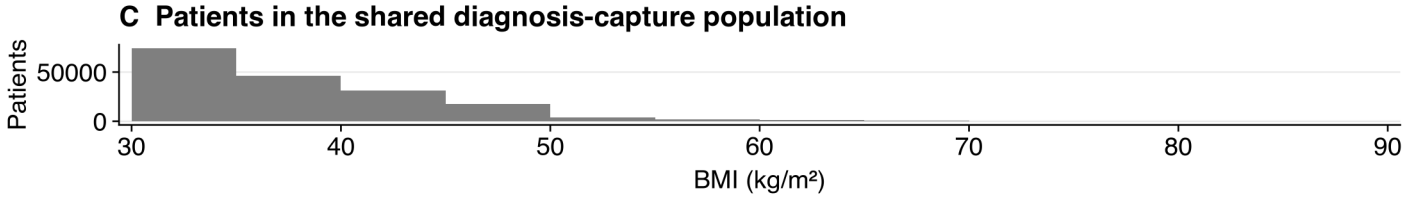
S2-ICD-specified

Sensitivity: exclude unspecified-only diagnosis category

Primary population | Encounter-wide; model/reference N = 176,794 | Maximum likelihood
Events: chronic/OHS 1,769; acute/chronic 3,443; acute 3,294; excluded 128

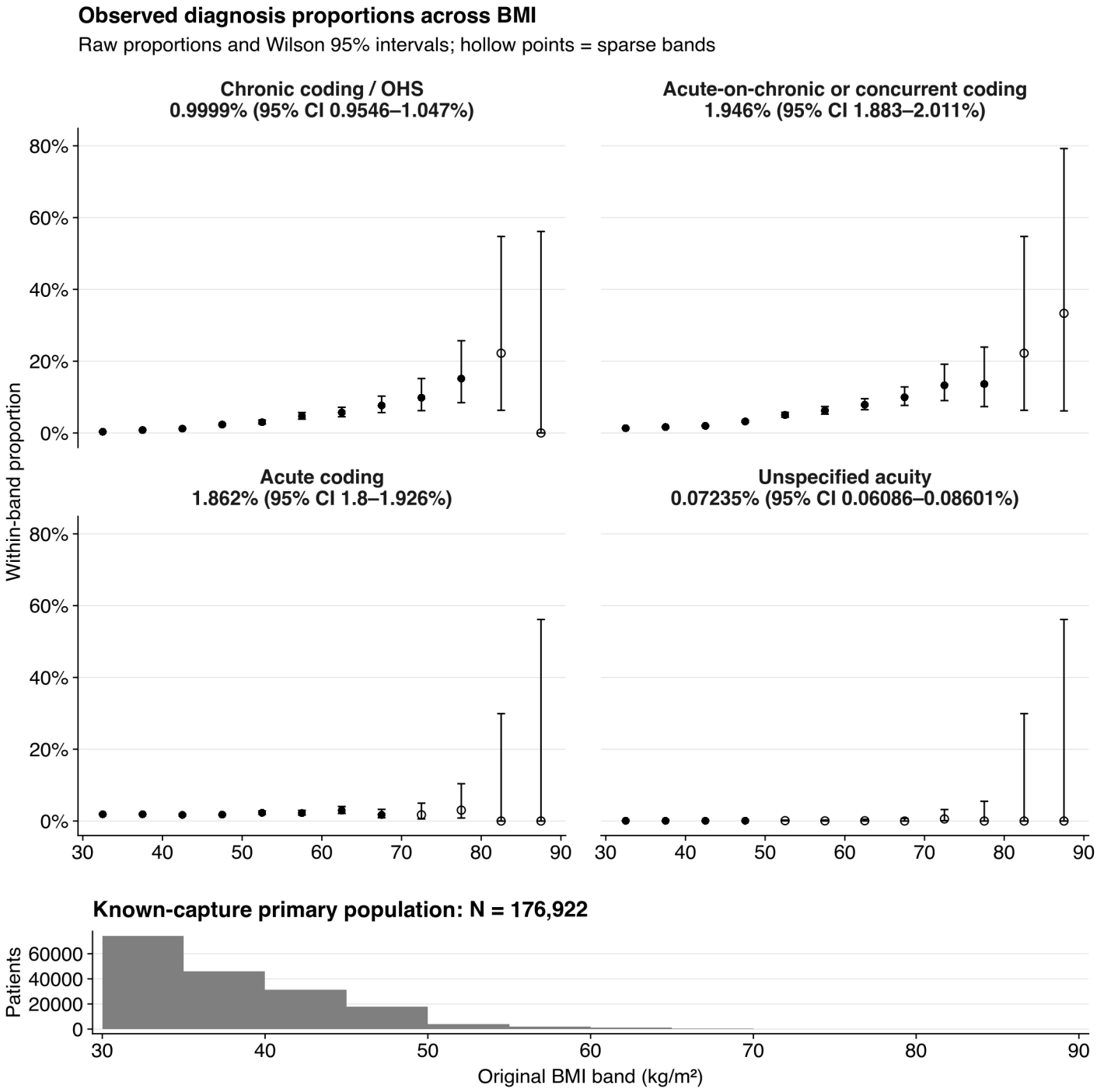






Unpenalized four-category sensitivity excluding unspecified-only encounters and standardizing over the remaining patients. Encounter-wide coding, no gas requirement; inherited adjustment and primary spline basis. Bands partition the composite; thickness gives component probability. Shading: pointwise 95% intervals from joint information covariance, conditional on the empirical reference and basis. Fading/dots: local sparsity; histogram N equals model/reference N. Acute-on-chronic includes explicit or concurrent acute/chronic coding. OHS (obesity hypoventilation syndrome) is analytically chronic; this does not establish chronic respiratory failure. Source selection and diagnostic incorporation limit physiological interpretation.

S2-ICD-observed



Observed encounter-wide diagnosis proportions within five-unit BMI bands of the covariate-complete primary population with known capture; no gas requirement. These are unadjusted descriptions, not fitted curves. Error bars are Wilson 95% intervals; hollow points mark sparse patient/event/non-event counts. The histogram shows each band denominator, shared by every category. Empty bands have no estimate. OHS is grouped with chronic coding analytically; codes do not establish physiology. Overall percentages and their Wilson intervals are printed in each panel; no-target coding is the complement of the four displayed categories.